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CSA1509-Cloud computing and Big Data Analytics

EXPERIMENT – 21

Running Virtual Machines with Different Configurations

Aim:

To run virtual machines with different configurations and determine how many virtual machines can be utilized simultaneously based on the available system resources.

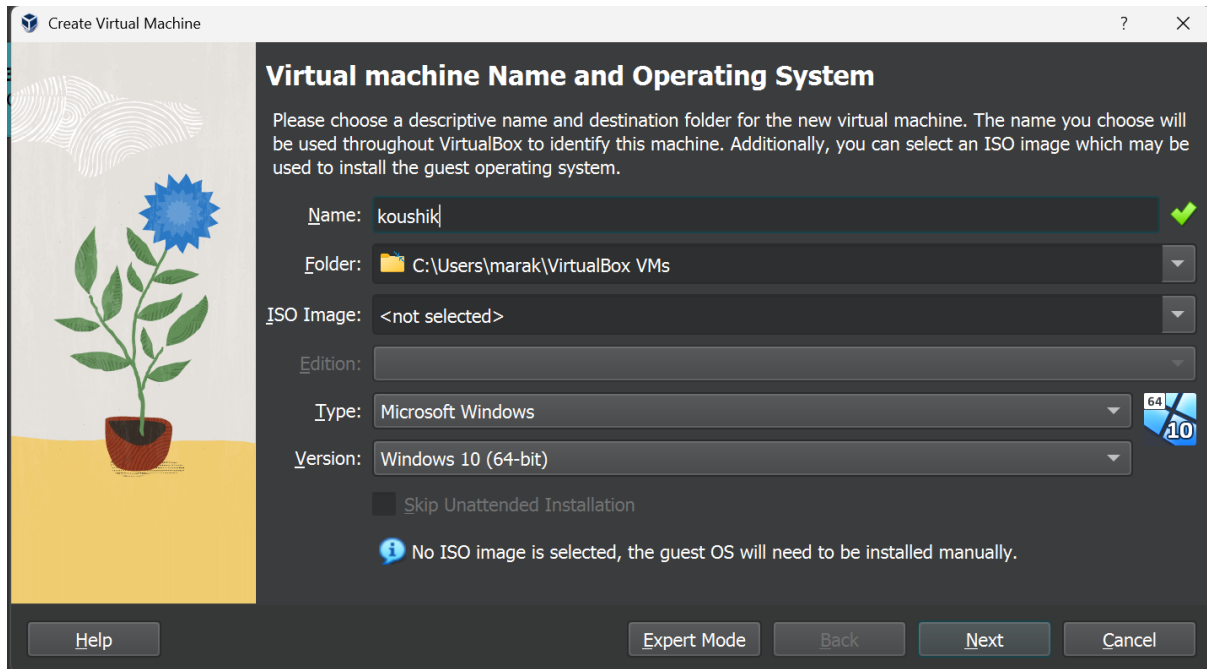
Short Procedure:

1. Open Oracle VirtualBox and create a new virtual machine.
2. Configure the VM with different RAM, CPU, and storage settings.
3. Install the required operating system in each virtual machine.
4. Start the virtual machines one by one and observe the system performance.
5. Continue running VMs until the available RAM/CPU resources are nearly utilized.
6. Record the number of VMs running simultaneously and their configurations.
7. Compare the configurations and determine the maximum number of VMs that can be utilized at a particular time.

Result:

The number of virtual machines that can run simultaneously depends on the host system's available RAM, CPU, and storage resources.

Step 1 : CREATING VM AND PROVIDING ISO FILE



Create Virtual Machine

Virtual machine Name and Operating System

Please choose a descriptive name and destination folder for the new virtual machine. The name you choose will be used throughout VirtualBox to identify this machine. Additionally, you can select an ISO image which may be used to install the guest operating system.

Name: ✓

Folder:

ISO Image:

Edition:

Type: 64

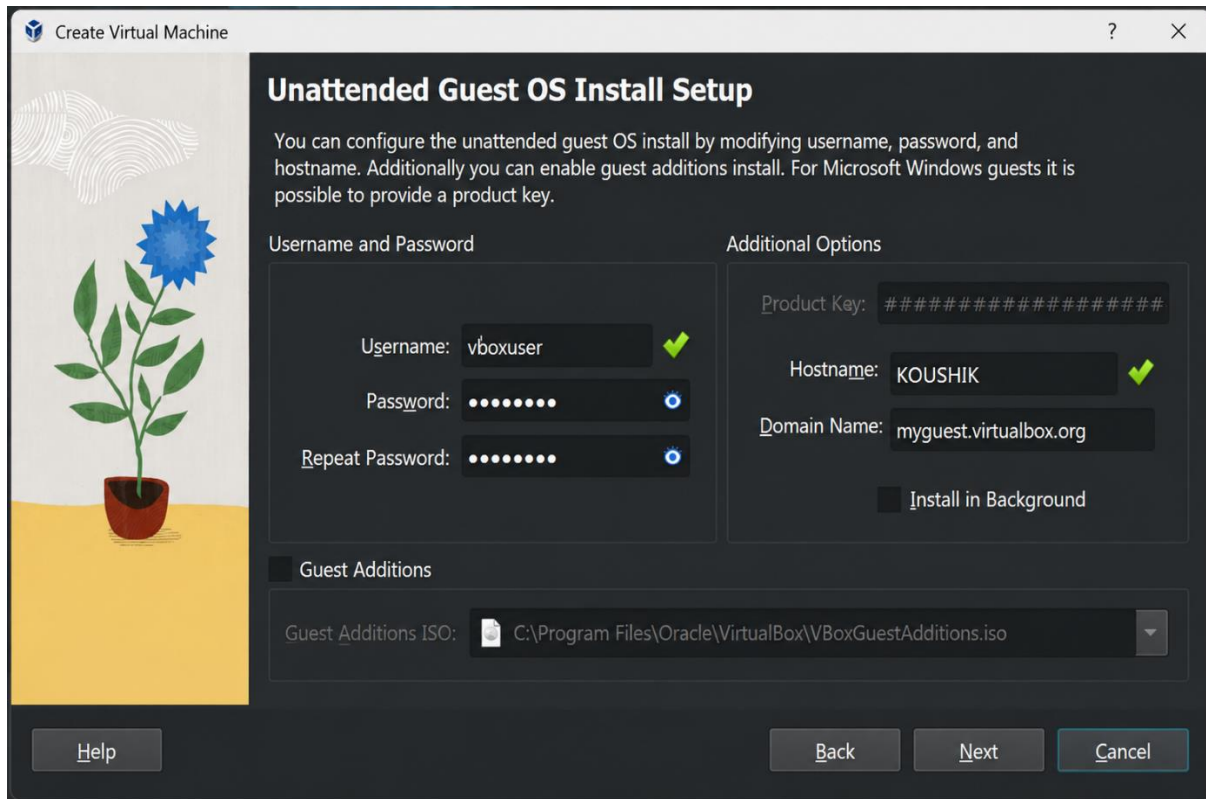
Version: 10

☐ Skip Unattended Installation

❗ No ISO image is selected, the guest OS will need to be installed manually.

Help Expert Mode Back Next Cancel

Step 2 : PROVIDING CREDENTIALS FOR THE VM



Create Virtual Machine

Unattended Guest OS Install Setup

You can configure the unattended guest OS install by modifying username, password, and hostname. Additionally you can enable guest additions install. For Microsoft Windows guests it is possible to provide a product key.

Username and Password

Username: ✓

Password: 🔑

Repeat Password: 🔑

Additional Options

Product Key:

Hostname: ✓

Domain Name:

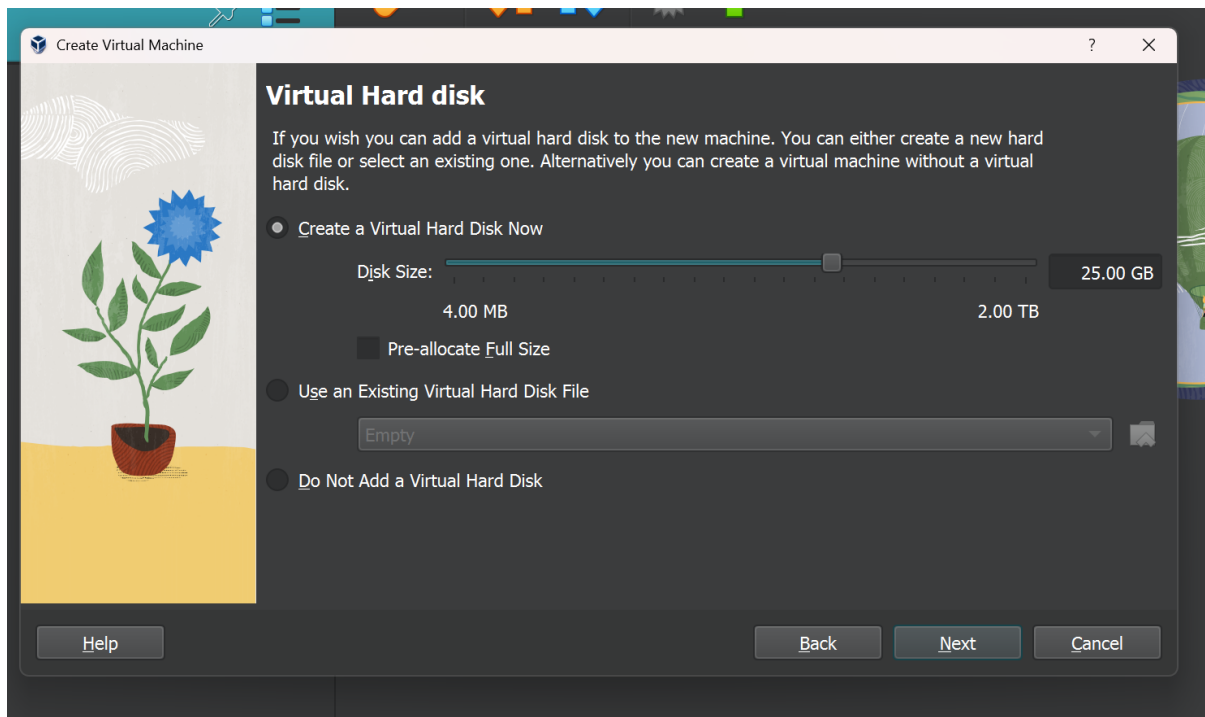
☐ Install in Background

Guest Additions

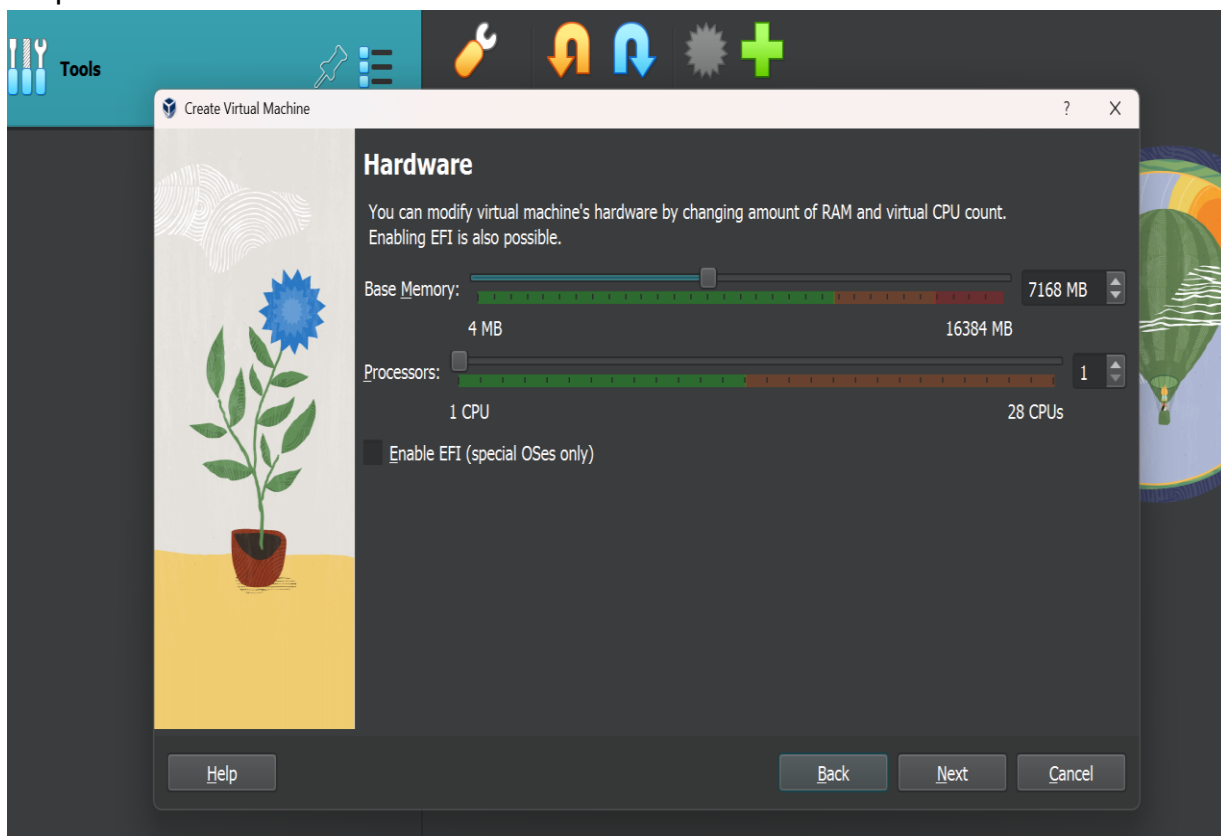
Guest Additions ISO:

Help Back Next Cancel

Step 3 : CREATING VIRTUAL HARD DISK WITH REQUIRED CONFIGURATION



Step 4 : CONFIGURING THE VM

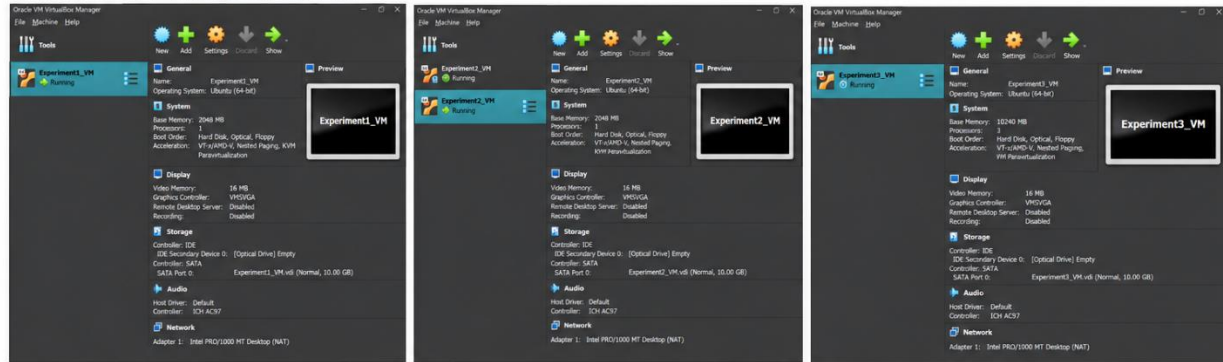


Step 5:- when take different configurations look like these

Record the number of VMs running simultaneously and their configurations.

VM Name	Operating System	RAM (MB)	CPU (Cores)	Storage (GB)	Status
Experiment1_VM	Ubuntu (64-bit)	2048	1	10.00	Running
Experiment2_VM	Ubuntu (64-bit)	2048	1	10.00	Running
Experiment3_VM	Ubuntu (64-bit)	10240	3	10.00	Running

VirtualBox Manager – Running Virtual Machines



Experiment1_VM (2 GB RAM, 1 CPU, 10 GB Storage)

Experiment2_VM (2 GB RAM, 1 CPU, 10 GB Storage)

Experiment3_VM (10 GB RAM, 3 CPU, 10 GB Storage)

Comparison and Maximum Number of VMs

Configuration	Total RAM Used (GB)	Total CPU Cores Used	Total Storage Used (GB)	No. of VMs Running		System Performance	Remarks
VM 1	2 GB	1	10 GB	1		Excellent	Smooth Performance
VM 1 + VM 2	4 GB	2	20 GB	2		Good	Slight Delay Observed
VM 1 + VM 2 + VM 3	14 GB	5	30 GB	3		Acceptable	High RAM Usage
Maximum VMs Running Simultaneously				3 VMs	14 GB RAM, 5 CPU Cores, 30 GB Storage		Beyond this system slows down

Conclusion: In this system configuration, a maximum of 3 virtual machines can be utilized simultaneously with acceptable performance.